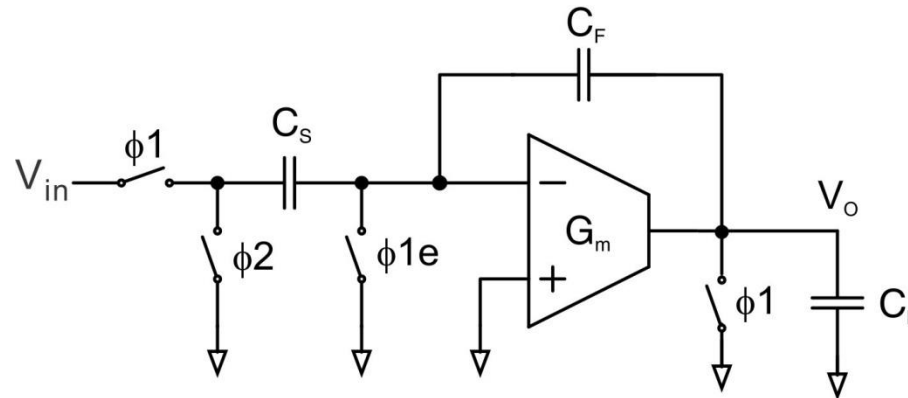


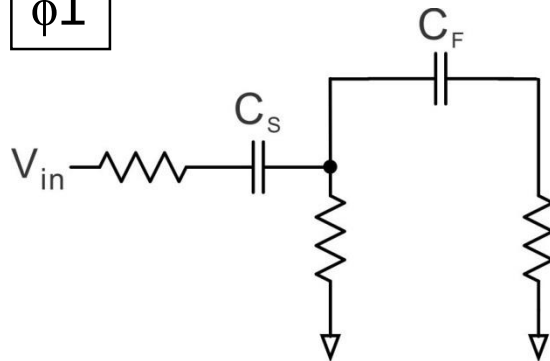
Switch Capacitor Circuit Settling and Noise Analysis

Nan Sun
Tsinghua University
nansun@Tsinghua.edu.cn

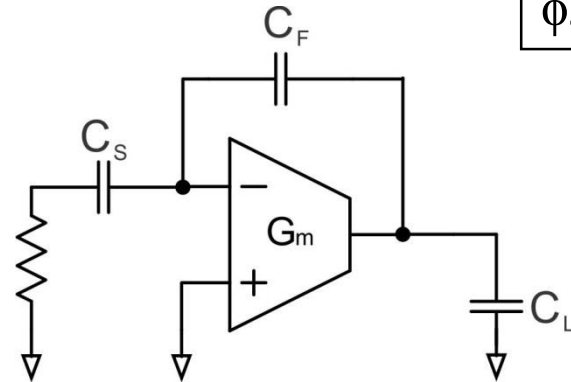
Settling Analysis



$\phi1$



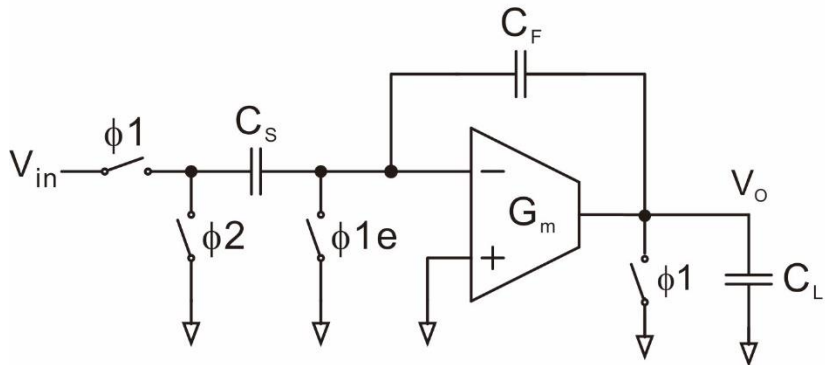
$\phi2$



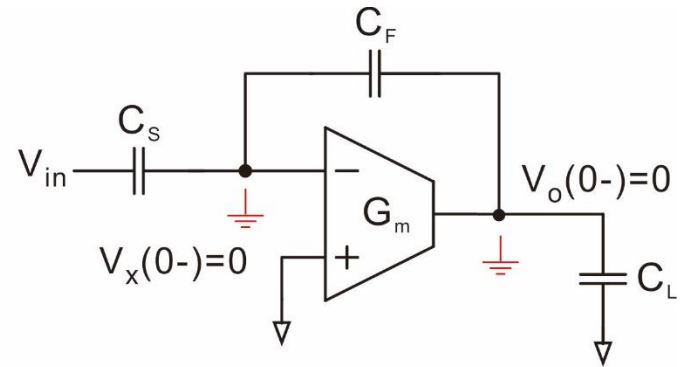
Ideal Case

- Let us assume the following ideal conditions
 - Ideal switch with no charge injection and clock feedthrough, and zero turn-on resistance
 - Ideal OTA with no input capacitance and infinite DC gain

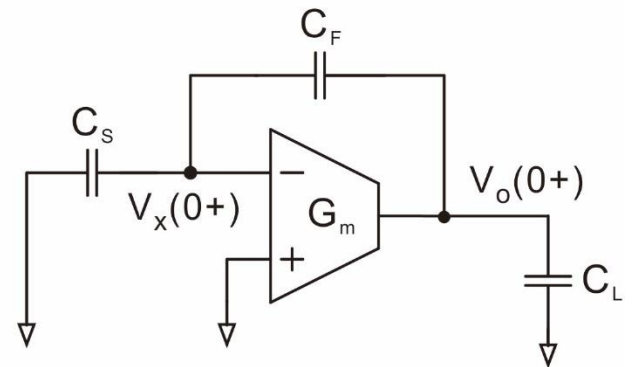
When ϕ_2 is on ($t=0$)



Right
before
 $\phi_2, t=0^-$



Right
After
 $\phi_2, t=0^+$



Solve for V_x and V_o at $t=0+$

- Charge conserves at V_o

$$Q_o(0-) = 0$$

$$Q_o(0+) = C_L V_o + C_F (V_o - V_x)$$

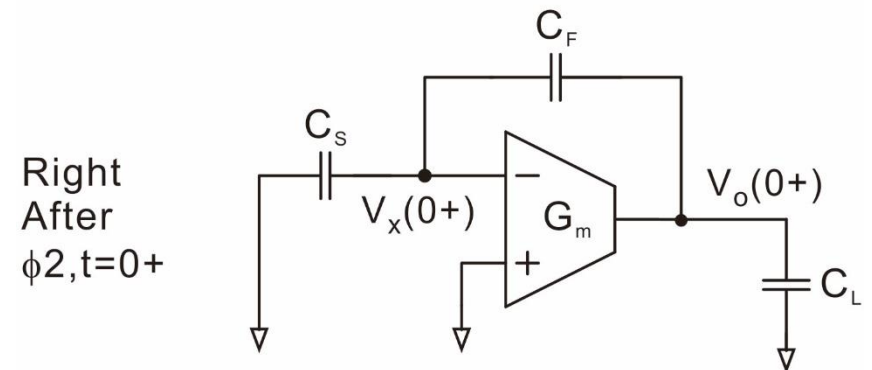
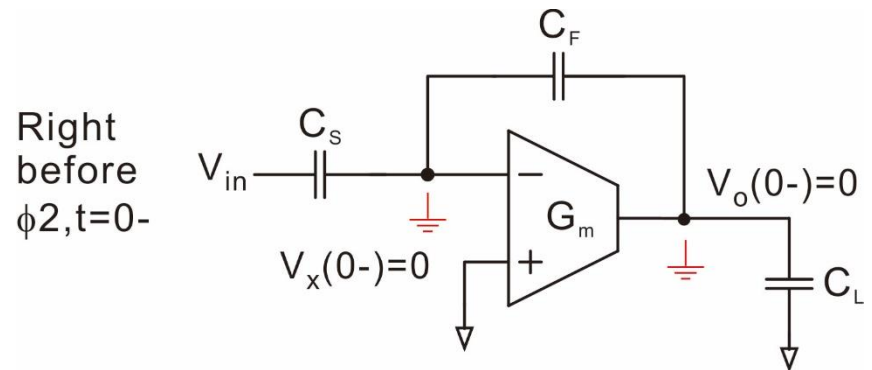
- Charge conserves at V_x

$$Q_x(0-) = -C_S V_{in}$$

$$Q_x(0+) = C_S V_x + C_F (V_x - V_o)$$

$$\Rightarrow V_x(0+) = -\frac{C_S}{C_S + C_F // C_L} V_{in}$$

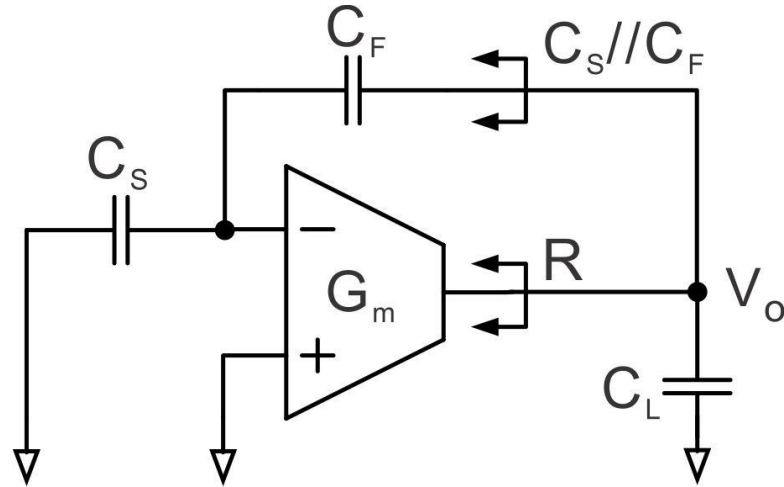
$$\Rightarrow V_o(0+) = \frac{C_F}{C_L + C_F} V_x(0+)$$



Time Constant

- First order settling during $\phi 2$

$$\beta = \frac{V_x}{V_o} = \frac{C_F}{C_F + C_S}$$



$$R = \frac{V_o}{g_m V_x} = \frac{1}{\beta \cdot g_m}$$

$$C_{Ltot} = C_L + (C_F // C_S)$$

$$\tau = RC_{Ltot} = \frac{C_{Ltot}}{\beta \cdot g_m}$$

Complete Expression for $t > 0$

$$V_X(t) = V_X(0+) \cdot e^{-t/\tau} + V_X(\infty) \cdot (1 - e^{-t/\tau}) = V_X(0+) \cdot e^{-t/\tau}$$

$$V_O(t) = V_O(0+) \cdot e^{-t/\tau} + V_O(\infty) \cdot (1 - e^{-t/\tau})$$

$$V_O(\infty) = \frac{C_S}{C_F} V_{in}$$

- Dynamic settling error decreases with time

$$e^{-t/\tau} < \varepsilon_d \quad \Rightarrow \quad N = \frac{t}{\tau} = -\ln \varepsilon_d$$

ε_d	N
1%	4.6
0.1%	6.9
0.01%	9.2

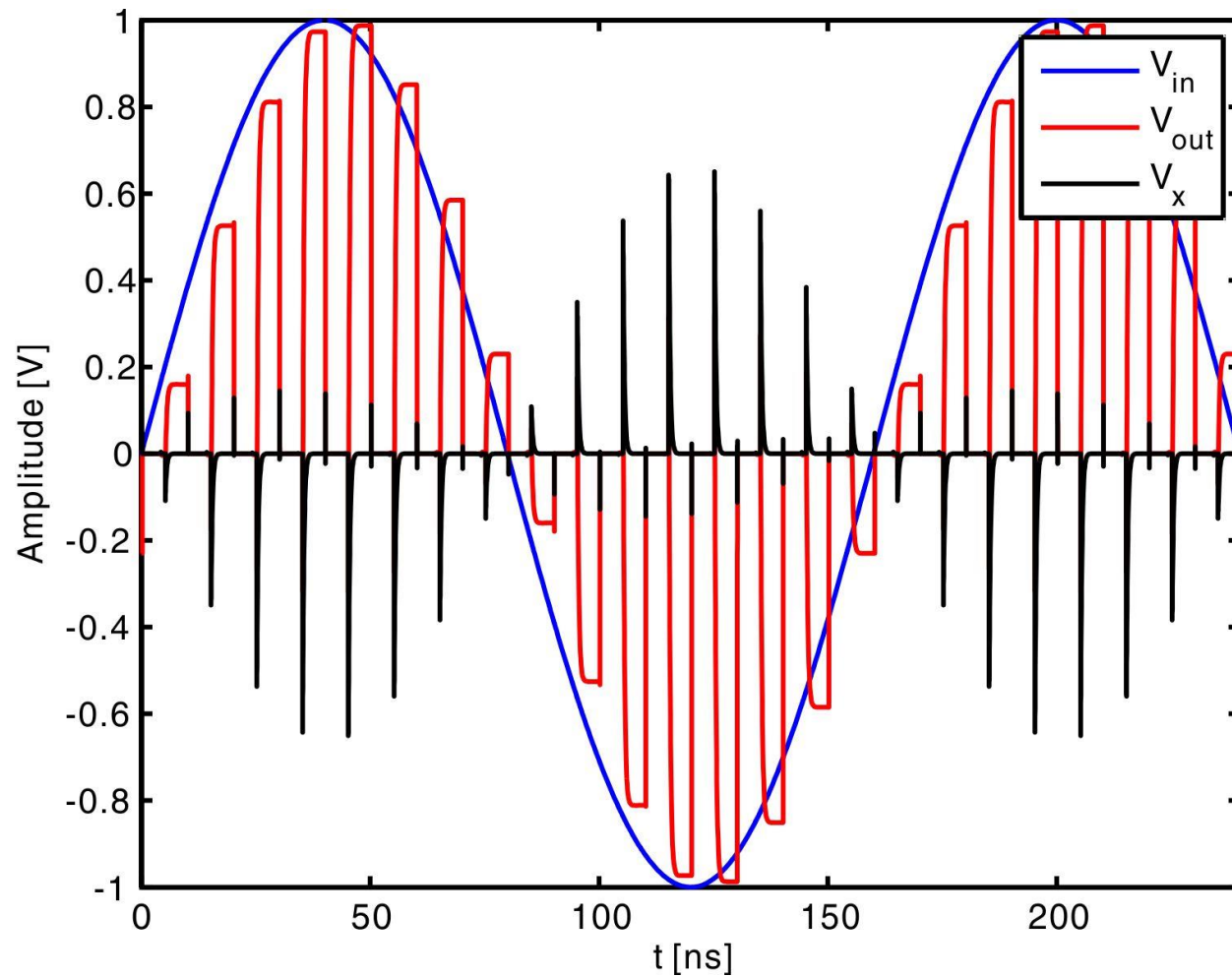
Example

- $C_S = C_F = C_L = 1\text{pF}$, $G_m = 10\text{mS}$, $R_{on} = 1\ \Omega$, and $f_s = 100\text{MS/s}$ ($t_s = 10\text{ns}$)

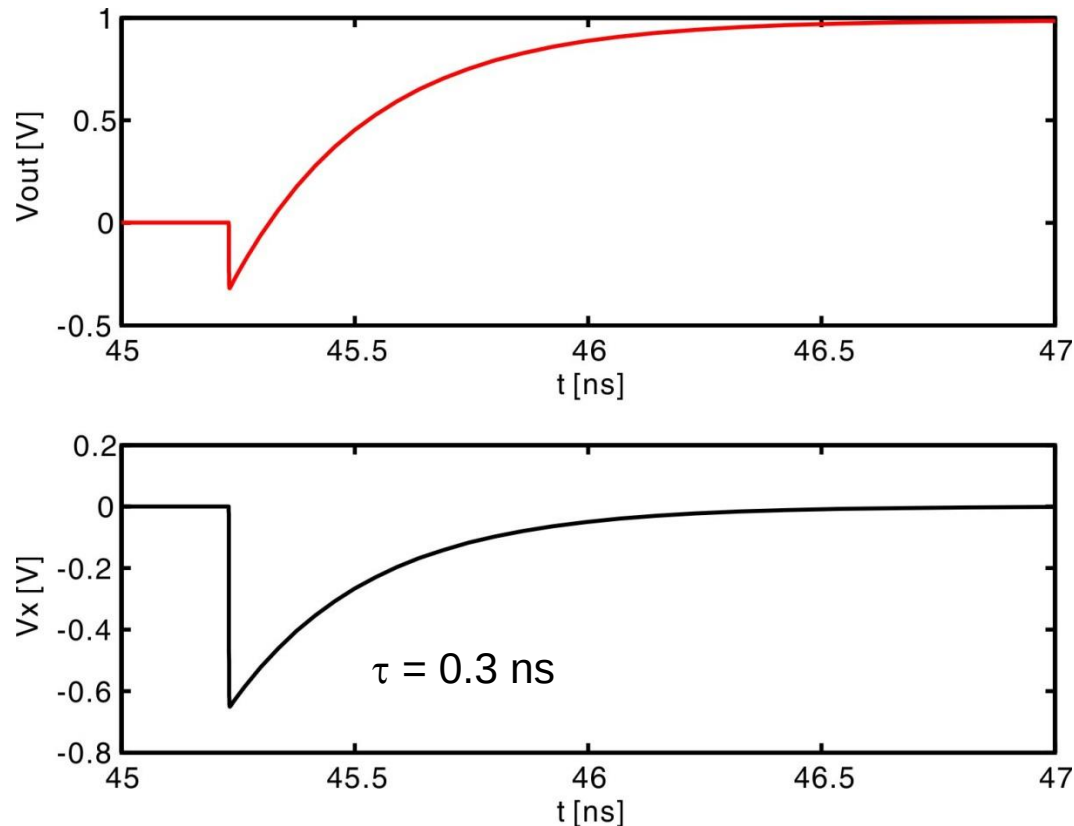
$$\beta = \frac{C_F}{C_F + C_S} = \frac{1}{2} \quad \tau = \frac{1}{\beta} \frac{C_{Ltot}}{g_m} = \frac{1.5\text{pF}}{0.5 \times 10\text{mS}} = 0.3\text{ns}$$

- We assume settling time = $t_s/2 \times 0.8 = 4\text{ns} \rightarrow N = 13$
- Does the dynamic error cause nonlinearity?
 - It does not cause nonlinearity for pure 1st-order linear settling
 - However, it is hard to ensure pure 1st-order settling
 - Initial slew
 - Switch has resistance, multiple poles and zeros
 - Signal-dependent charge injection
 - Clock jitter

Complete Settling with $G_m = 10\text{mS}$

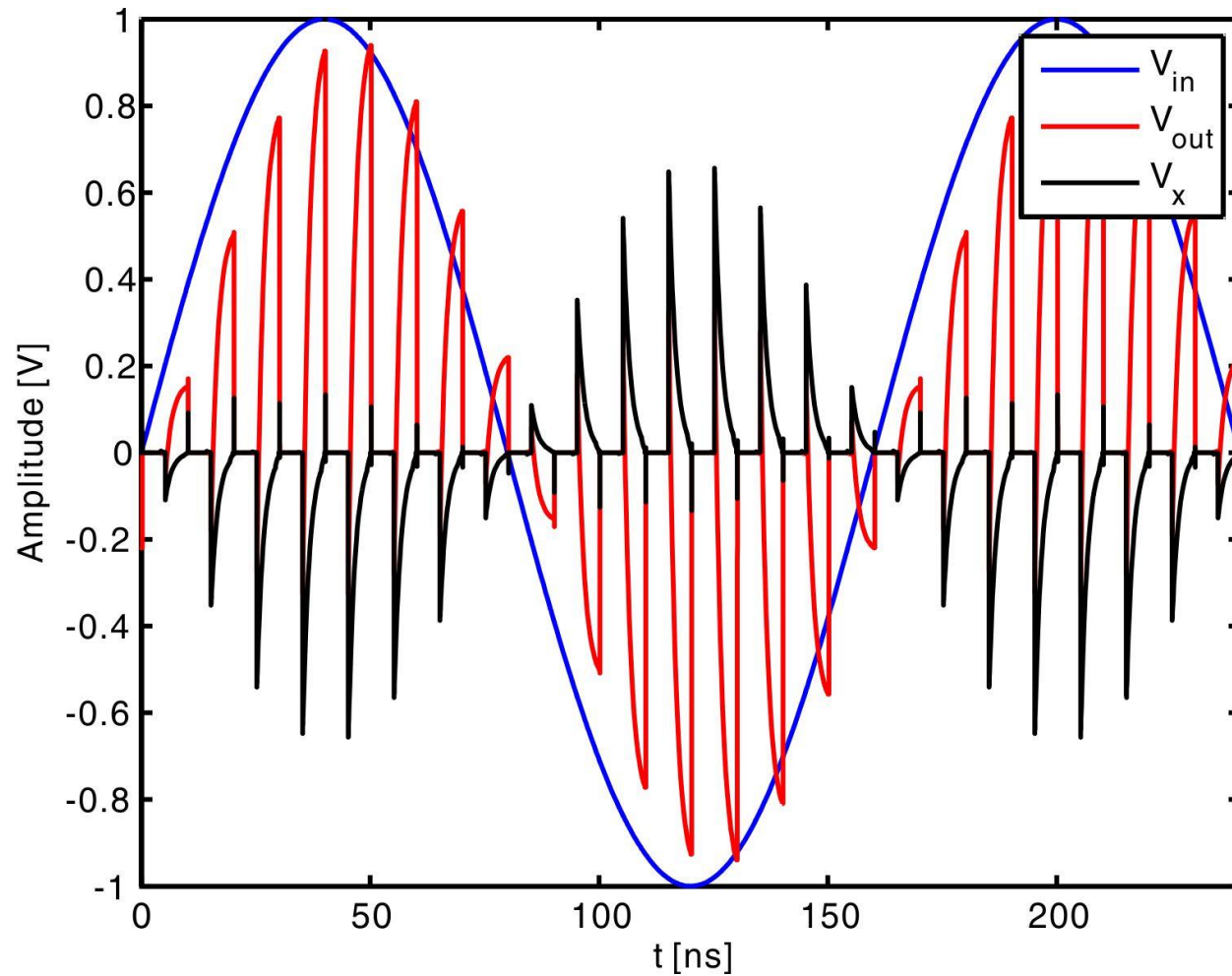


Zoom in around $V_{in} = 1V$

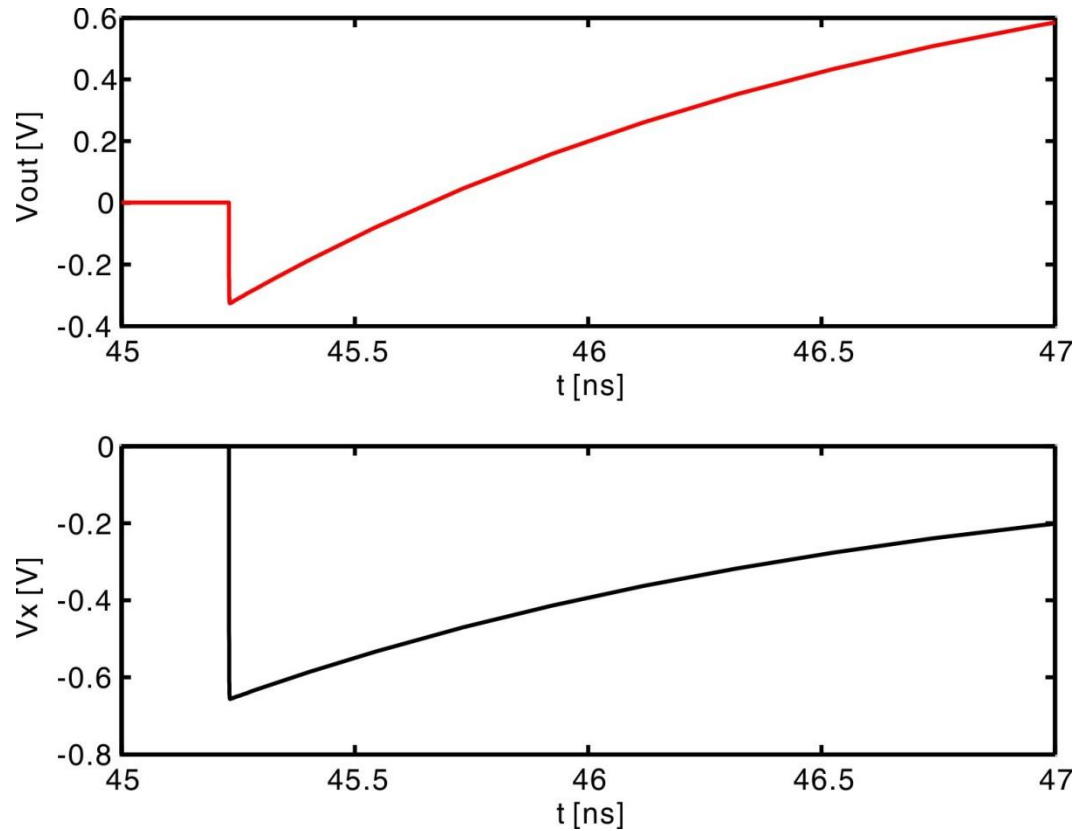


- $V_x(0-) = V_{out}(0-) = 0$
- $V_x(0+) = -(1\text{pF}/1.5\text{pF})V_{in} = -0.66V$, $V_{out}(0+) = (1\text{pF}/2\text{pF})V_x(0+) = -0.33V$
- Be careful about the droop at $V_x \rightarrow$ it may cause the OTA to turn off

Incomplete Settling with $G_m = 2\text{mS}$



Zoom in around $V_{in} = 1$



- $\tau = 1.5$ ns
- Settling is incomplete

Nonidealities

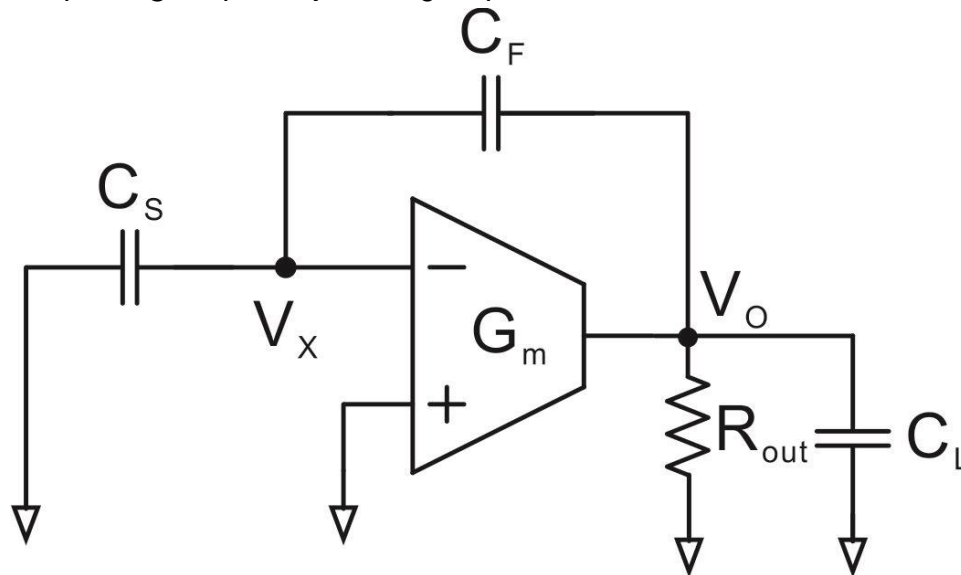
- Parasitic capacitances
- Finite OTA gain
- Finite OTA output current (slew-rate limited)

Parasitic Capacitance at V_x

- Parasitic capacitance at V_x node (the OTA input)
 - Bottom-plate capacitance
 - Switch capacitance
 - OTA input capacitance
- Effects
 - Reduced feedback factor $\beta = C_F / (C_S + C_F + C_X)$
 - Increased loading $C_{Ltot} = C_L + C_F // (C_S + C_X)$
 - Slower settling

Finite OTA Gain

- OTA has finite gain $A_V = G_m R_{out}$
- $V_X = -V_O/A_V \rightarrow$ not virtual ground $\rightarrow$ static settling error
- $Q_X = -C_S V_{in} = C_S V_X + C_F (V_X - V_O) = -V_O [C_F + (C_S + C_F)/A_V]$
- $\rightarrow V_O/V_{in} = C_S / [C_F + (C_S + C_F)/A_V] = C_S / C_F [T / (1 + T)]$ where $T = \beta A_V$



Static Settling Error (1)

- Ideal output voltage for $t \rightarrow \infty$

$$V_{o,ideal} = V_{in} \frac{C_S}{C_F}$$

- Actual output voltage

$$V_o = V_{in} \frac{C_S}{C_F} \frac{T}{1+T} = V_{o,ideal} \frac{T}{1+T} \quad T = \beta \cdot A_v$$

- Define static settling error ε_s

$$\varepsilon_s = \frac{V_o - V_{o,ideal}}{V_o} = -\frac{1}{T}$$

- E.g., $A_v = 1000$, $T = 500$, $\rightarrow 0.2\%$ static settling error

Static Settling Error (2)

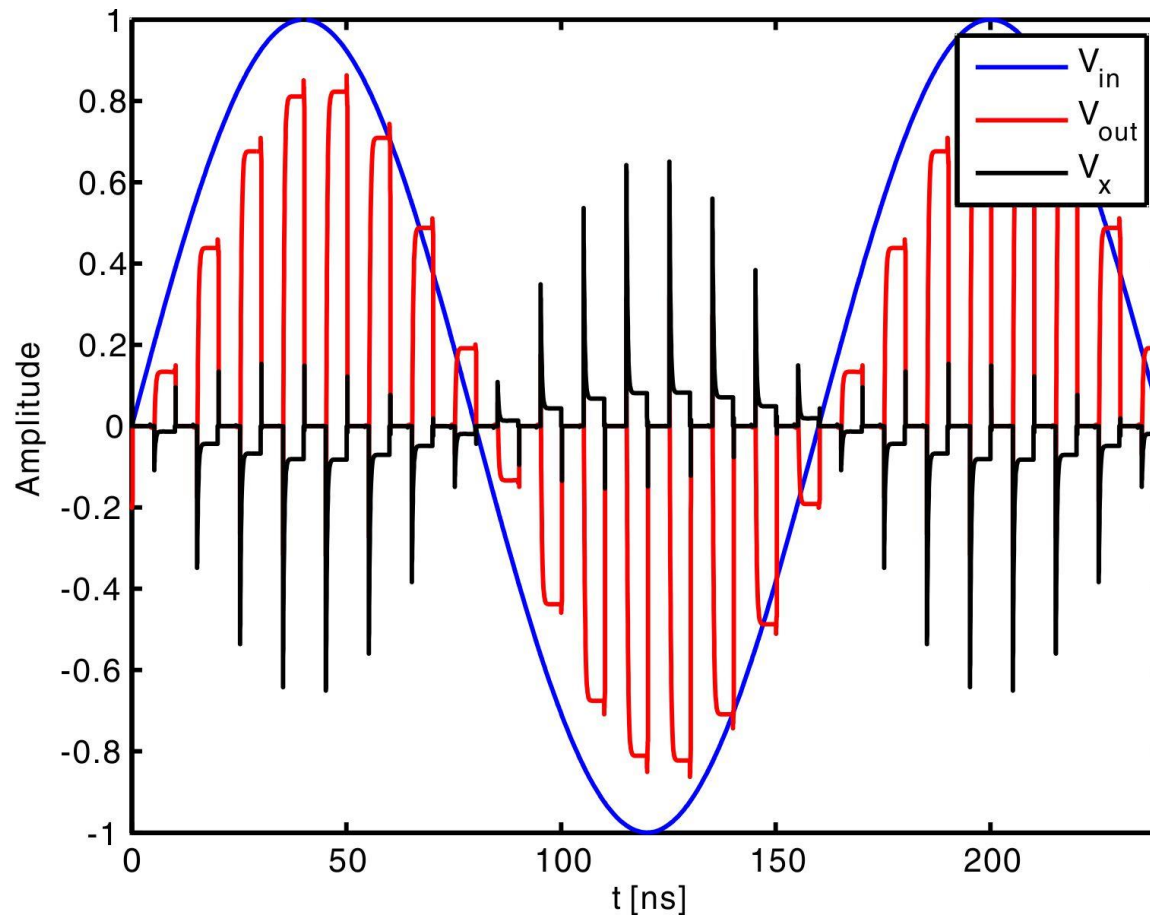
□ $\varepsilon_s = -1/T$

ε_s	T
<1%	> 40dB
<0.1%	> 60dB
<0.01%	> 80dB

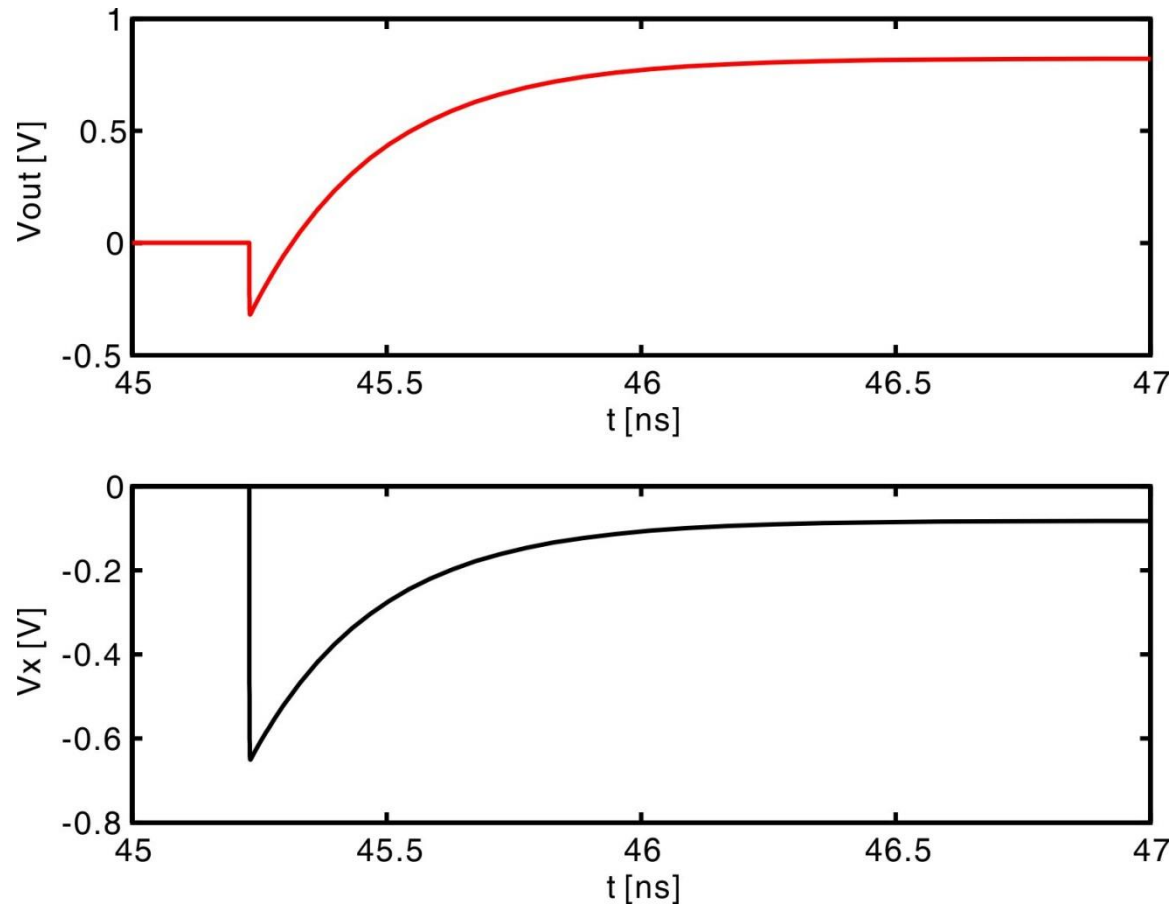
- Does the static error cause nonlinearity?
 - It does not cause nonlinearity assuming a constant A_v
 - However, A_v is not a constant, but signal dependent
 - Input-dependent A_v causes nonlinearity
 - The common design practice is to make A_v so large that the static error is small and the nonlinearity is negligible

Insufficient Loop Gain

- E.g., $A_v = 10$, $T = 5$

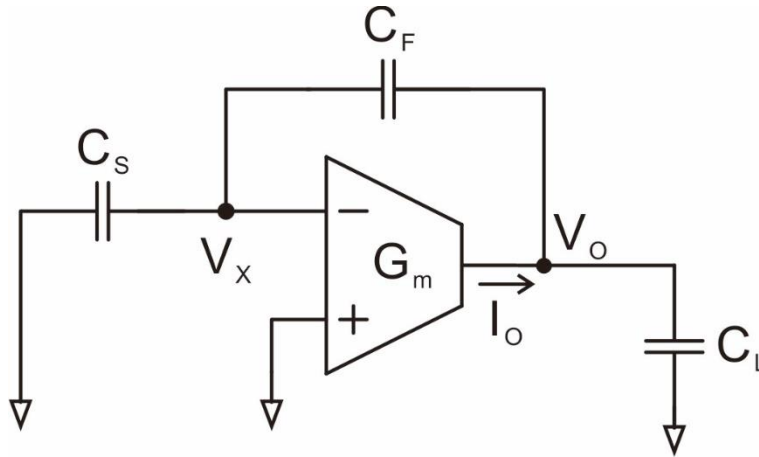


Zoom in around $V_{in} = 1$



- V_{out} and V_x settle to 0.082 and 0.82 respectively $\rightarrow$ 20% static settling error

Finite OTA Output Current



$$V_o(t) = V_o(\infty) + [V_o(0+) - V_o(\infty)]e^{-t/\tau}$$

- The current delivered by the OTA is

$$I_o(t) = -C_{Ltot} \frac{dV_o}{dt} = -C_{Ltot} \frac{V_o(0+) - V_o(\infty)}{\tau} e^{-t/\tau}$$

- Peak current occurs at $t=0$

$$|I_{O,max}| = C_{Ltot} \frac{|V_o(0+) - V_o(\infty)|}{\tau}$$

Slewing

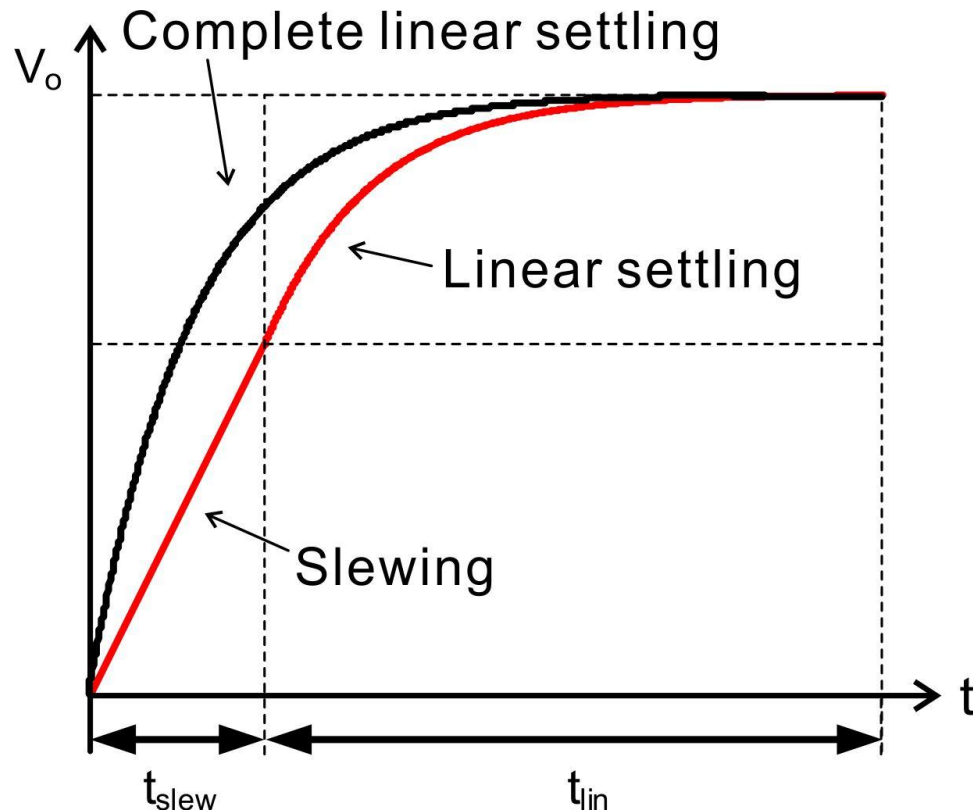
- Assuming an OTA can at most deliver I_D
 - No slewing means if $I_{o,\max} < I_D$

$$|I_{O,\max}| = C_{Ltot} \frac{|V_O(0+) - V_O(\infty)|}{\tau} < I_D$$

$$C_{Ltot} \frac{|V_O(0+) - V_O(\infty)|}{\frac{1}{\beta} \frac{C_{Ltot}}{g_m}} < I_D \quad \Rightarrow \quad \frac{g_m}{I_D} < \frac{1}{\beta |V_O(0+) - V_O(\infty)|}$$

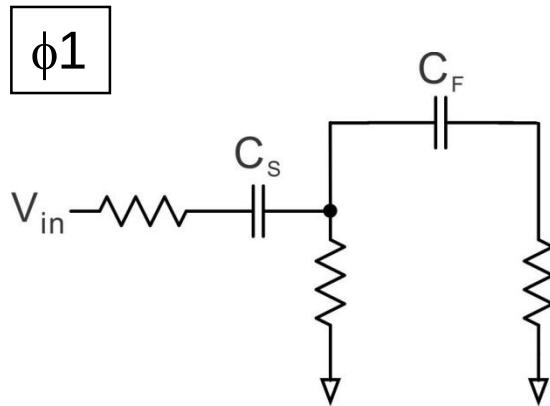
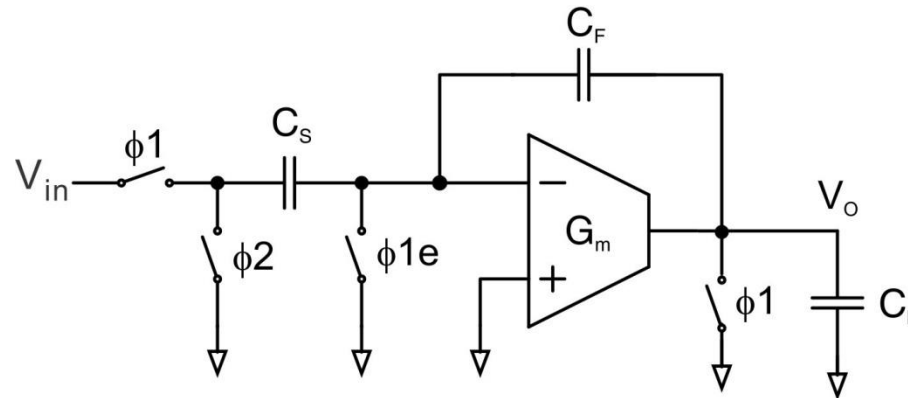
- Example: $\beta=0.5$, $|V_O(0+) - V_O(\infty)| = 1V \rightarrow g_m/I_D < 2 \text{ S/A}$
- Low g_m/I_D means low power efficiency
- To increase allowed g_m/I_D
 - Small output swing $\rightarrow$ hurt dynamic range
 - Small $\beta \rightarrow$ large closed-loop gain $\rightarrow$ smaller input signal swing

Output Waveform with Initial Slewing

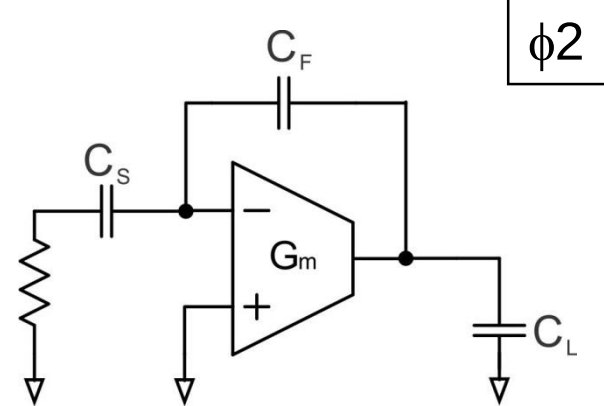


- Settling will be slower with slewing
- The dynamic static error causes nonlinearity

Noise Analysis

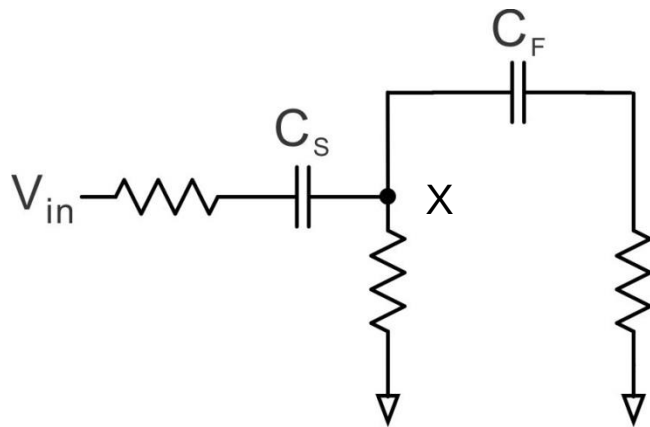


Noise due to switches



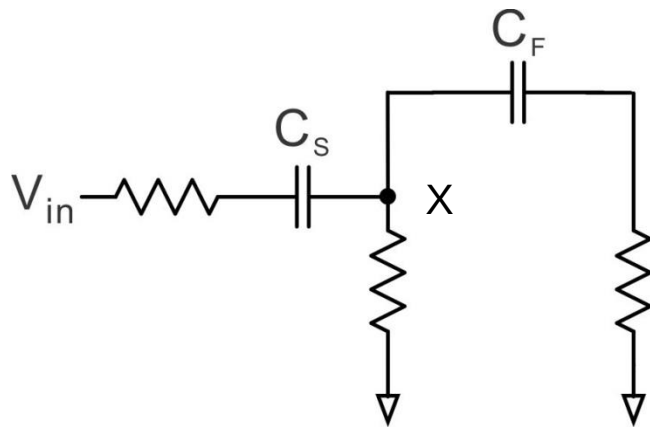
Noise due to OTA and switches

Sampling Phase ($\phi 1$)



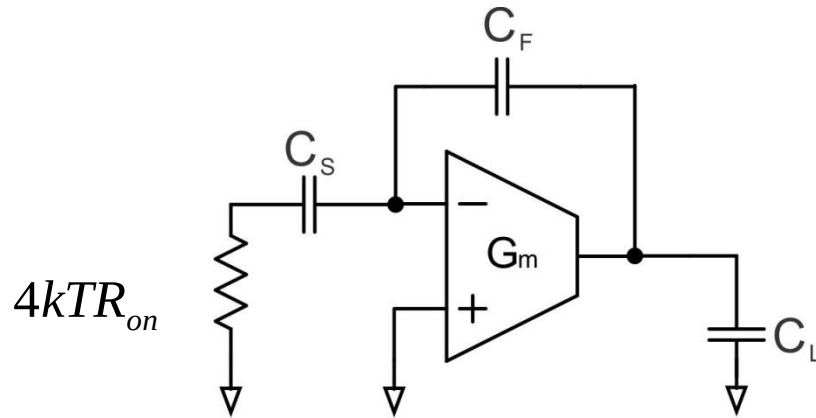
- Variable of interest is total "noise charge" at node X , q_n^2
- Cumbersome to compute using standard analysis
 - Find transfer function from each noise source (3 resistors) to q_x
 - Integrate magnitude squared expressions from zero to infinity and add
- Use equipartition theorem
 - Much easier 😊

Sampling Phase Noise Charge



- Noise charge at right plate of C_s
 - $q_{ns}^2 = kT \times C_s$
- Noise charge at the left plate of C_f
 - $q_{nf}^2 = kT \times C_f$
- Total noise charge at node X:
 - $q_n^2 = kT (C_s + C_f)$
- Note that any additional parasitic capacitance at node X will increase the sampled noise charge!

Charge Transfer Phase (ϕ_2)

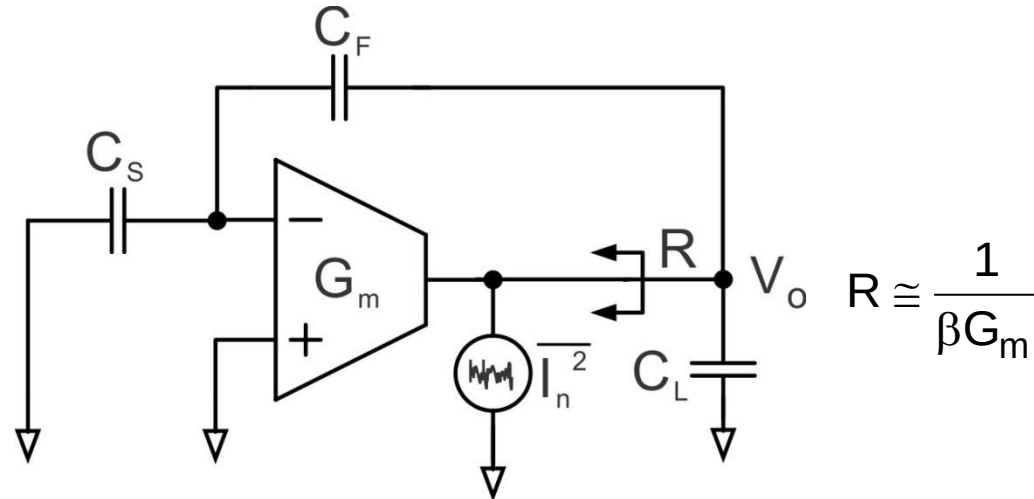


$\alpha > 1$ excess noise factor

$$\overline{V_n^2} / \Delta f = 4kT \frac{\alpha \gamma}{G_m}$$

- In a proper design, R_{on} is much smaller than $\alpha \gamma / G_m$
 - The noise should be dominated by OTA
 - It is much easier to design switches with low on-resistance than an amplifier with very large G_m
 - Switch power is much smaller than OTA

Total Integrated Amplifier Noise



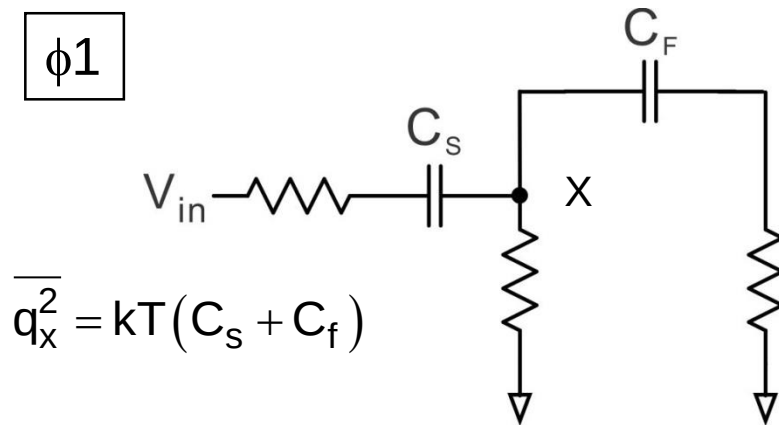
$$\overline{I_n^2 / \Delta f} = \alpha \cdot 4kT\gamma G_m$$

$$\frac{\overline{v_o^2}}{\Delta f} = \alpha \cdot 4kT\gamma \frac{1}{\beta R} \cdot \left| R \parallel \frac{1}{j\omega C_{Ltot}} \right|^2$$

$$\overline{v_o^2} = \int_0^\infty \alpha \cdot 4kT\gamma \frac{1}{\beta R} \cdot \Delta f \cdot \left| \frac{R}{1 + j\omega RC_{Ltot}} \right|^2 df = \alpha\gamma \frac{1}{\beta} \frac{kT}{C_{Ltot}}$$

Adding up Noise Contributions

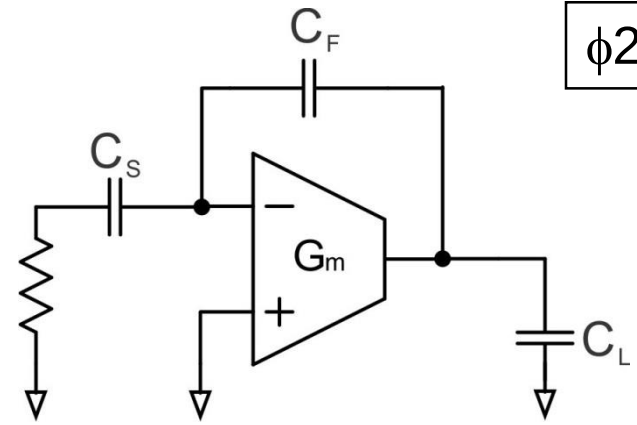
$\phi 1$



$$\overline{q_x^2} = kT(C_s + C_f)$$

$$\overline{v_{o,1}^2} = \frac{\overline{q_x^2}}{C_f^2} = kT \left(\frac{C_s + C_f}{C_f^2} \right) = \frac{kT}{C_f} \left(1 + \frac{C_s}{C_f} \right)$$

$\phi 2$



$$\overline{v_{o,2}^2} \cong \alpha \gamma \frac{1}{\beta} \frac{kT}{C_{Ltot}}$$

$$\overline{v_{o,tot}^2} = \frac{kT}{C_f} \left(1 + \frac{C_s}{C_f} \right) + \alpha \gamma \frac{1}{\beta} \frac{kT}{C_{Ltot}}$$

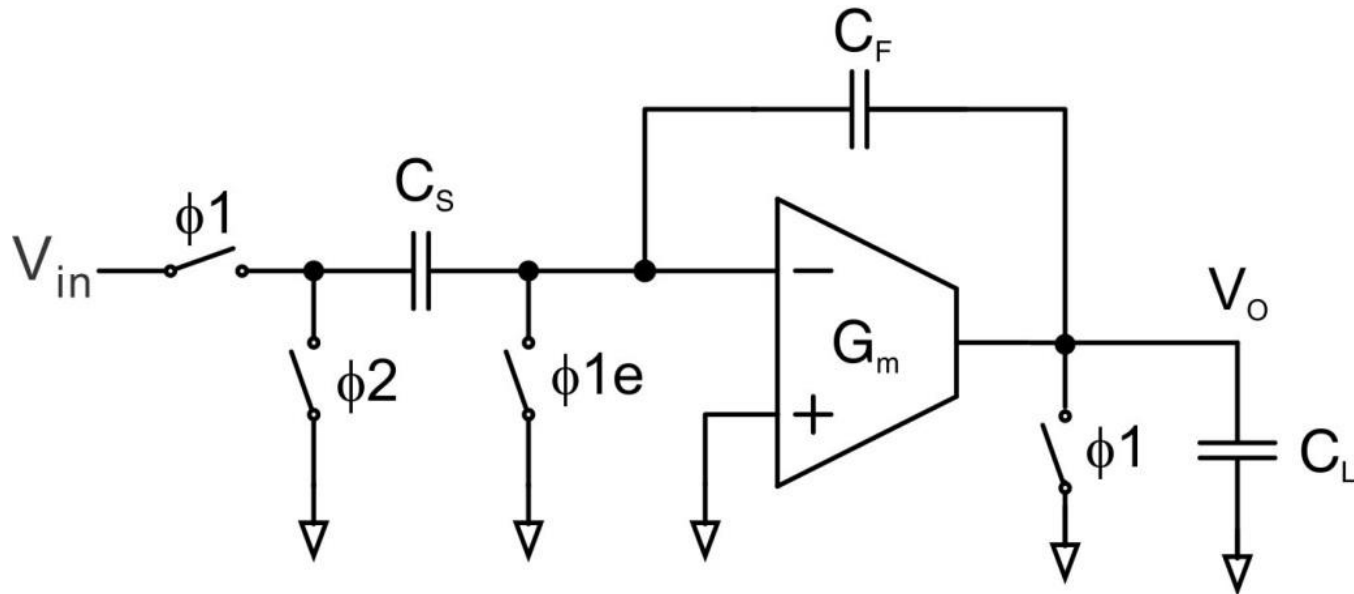
Noise in Differential Circuits

- In differential circuits, the noise power is doubled (because there are two half circuits contributing to the noise)
- But, the signal power increases by 4x
 - Looks like a 3dB win?
 - $\text{SNR}_{\text{single}} \propto P_{\text{sig}}/P_{\text{N}}$
 - $\text{SNR}_{\text{diff}} \propto 4P_{\text{sig}}/2P_{\text{N}} = 2P_{\text{sig}}/P_{\text{N}}$
- There is a 3dB win in DR, but it comes with the price of twice the power dissipation (due to two half circuits)
- Can get the same DR/power in a single ended circuit by doubling all cap sizes and g_{m}
- So what is the benefit of using fully-differential circuit?

Noise Simulation Example

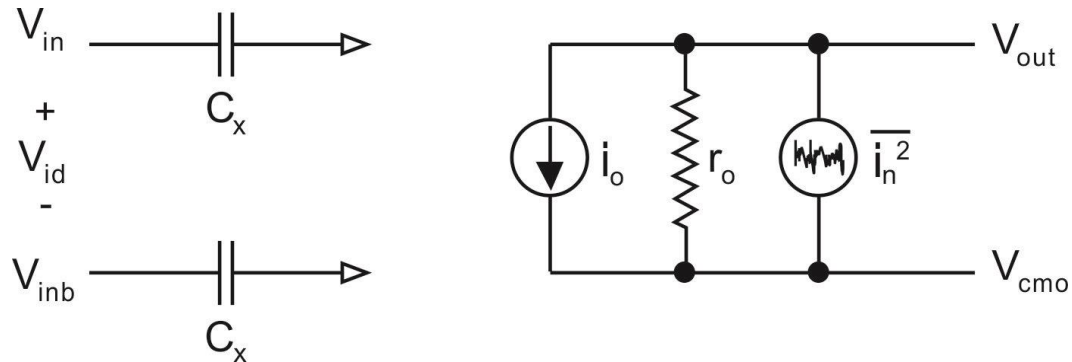
- Three ways to simulate noise in switched capacitor circuits
- Basic .ac/.noise Spice simulations
 - Must simulate noise in each clock phase separately
 - Activate $\phi 1$ switches, run .noise and integrate noise charge at relevant node over all frequencies and refer to output
 - Activate $\phi 2$ switches, run .noise and integrate noise at output
- Periodic Steady State Simulation
 - E.g. SpectreRF or BDA, “periodic noise analysis” (PNOISE)
 - Allows to simulate noise while switched capacitor circuit is clocked between $\phi 1$ and $\phi 2$
 - Noise from all phases is automatically added, all correlation taken care of
- Transient Noise

Example Track-and-Hold



Parameters: $f_s = 100\text{MHz}$, $R_{on} = 100\Omega$, $C_S = C_F = 1\text{pF}$,
 $C_L = 1\text{pF}$

OTA Simulation Model



$$C_x = \frac{g_m}{2\pi f_T}$$

$$\frac{\overline{I_n^2}}{\Delta f} = 8kTg_m$$

$$r_o = \frac{A_v}{g_m}$$

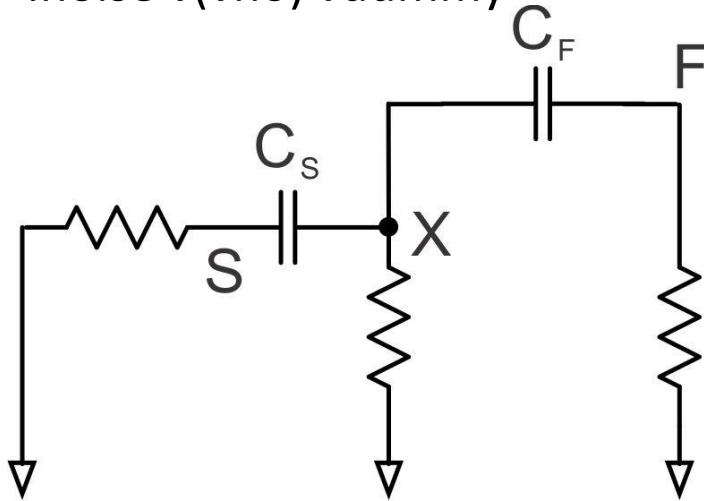
Parameters: $g_m=5\text{mS}$, $A_v=1\text{e}3$, $g_m/I_D = 10\text{S/A}$,
 $f_T=20\text{GHz}$

Sampling Phase (.noise)

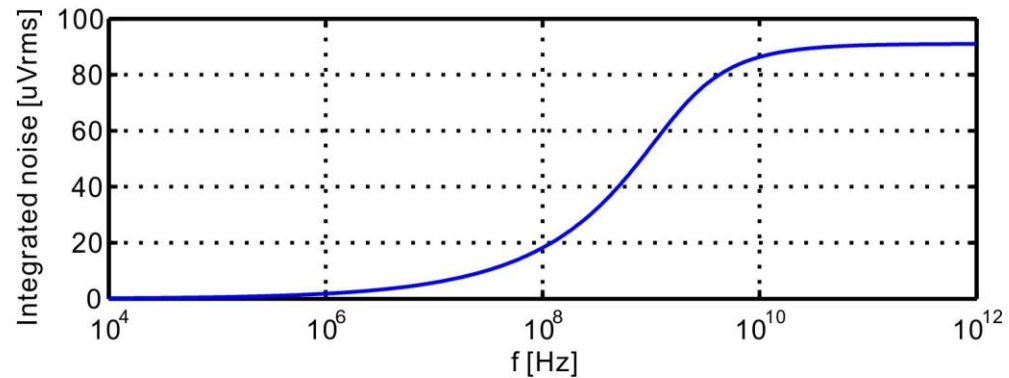
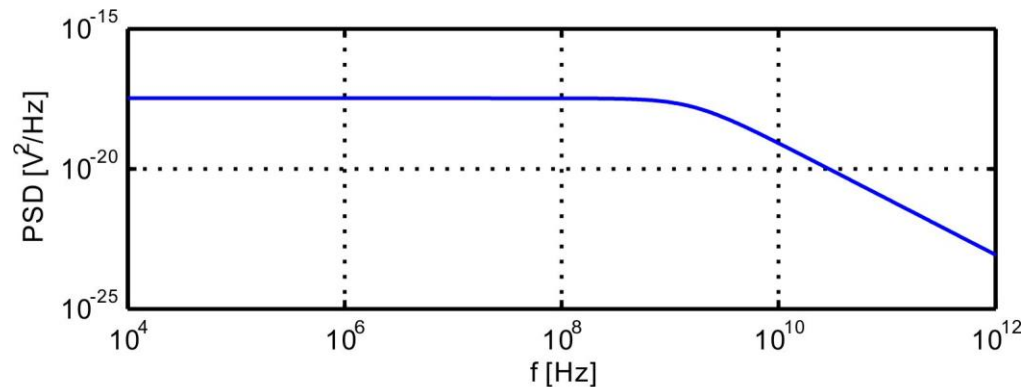
HSPICE: `en vno 0 vcvs vol=(cs*v(x,s) + cf*v(x,f))/cf`

`.ac dec 100 10k 1T`

`.noise v(vno) vdummy`

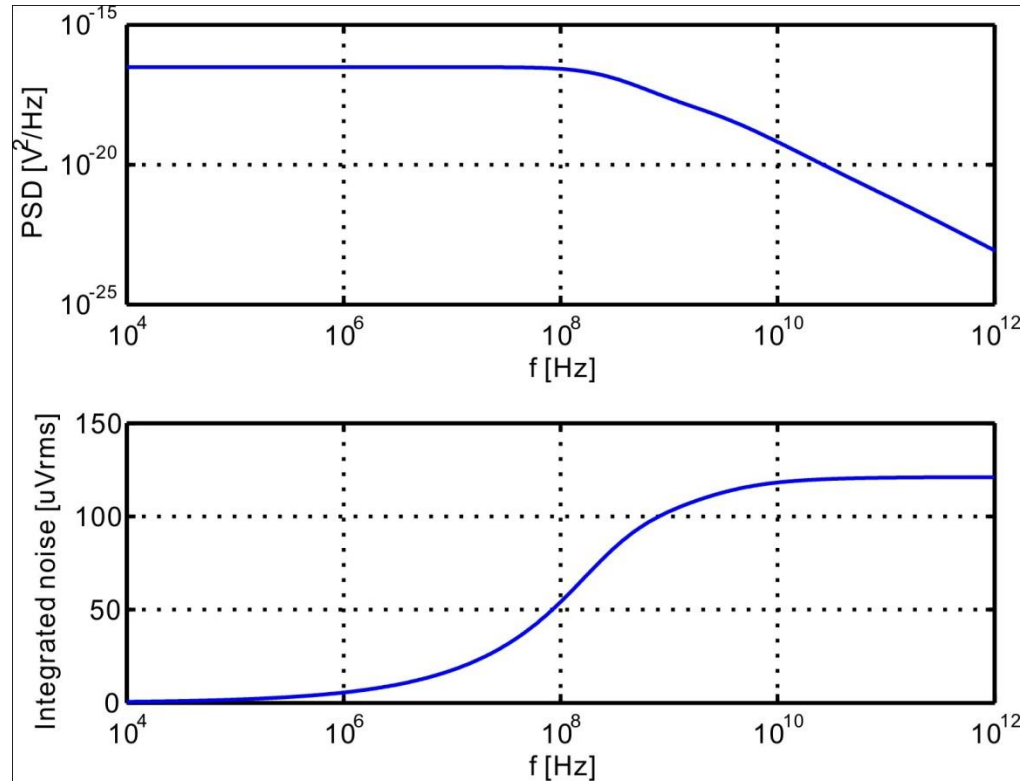
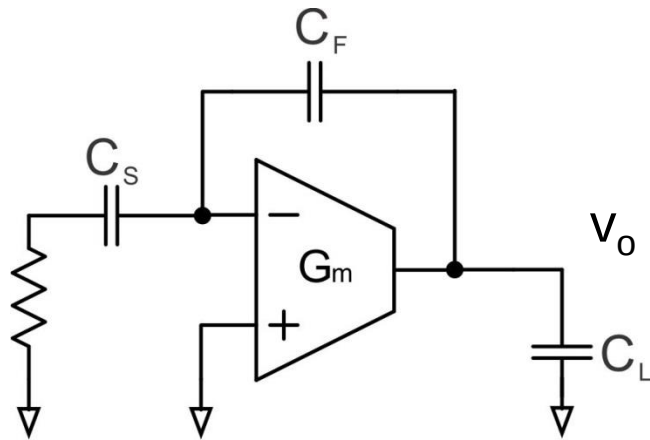


In Cadence, use `vcvs` in `analogLib` to add multiple noise components



Hand calculation: $91\mu V_{rms}$ ($64\mu V \times 1.414$)

Charge Transfer Phase (.noise)



Hand calculation : 104 μV_{rms} ($\alpha\gamma/\beta=4$, $C_{Leff}=1.5pF$)

PSS Setup

Analysis

☐ tran ☐ dc ☐ ac ☐ noise

☐ xf ☐ sens ☐ dcmatch ☐ stb

☐ pz ☐ sp ☐ envlp ☒ pss

☐ pac ☐ pstb ☐ pnoise ☐ pxf

☐ psp ☐ qpss ☐ qpac ☐ qpnoise

☐ qpxf ☐ qpss

Periodic Steady State Analysis

Engine ☒ Shooting ☐ Flexible Balance

Fundamental Tones

#	Name	Expr	Value	Signal	SrcId
---	------	------	-------	--------	-------

Clear/Add Delete Update From Schematic

☒ Beat Frequency ☐ Beat Period 100M Auto Calculate ☐

Output harmonics

Number of harmonics 0

Accuracy Defaults (errpreset)

☐ conservative ☒ moderate ☐ liberal

Additional Time for Stabilization (tstab) 0

Save Initial Transient Results (saveinit) ☐ no ☐ yes

Oscillator ☐

Sweep ☐

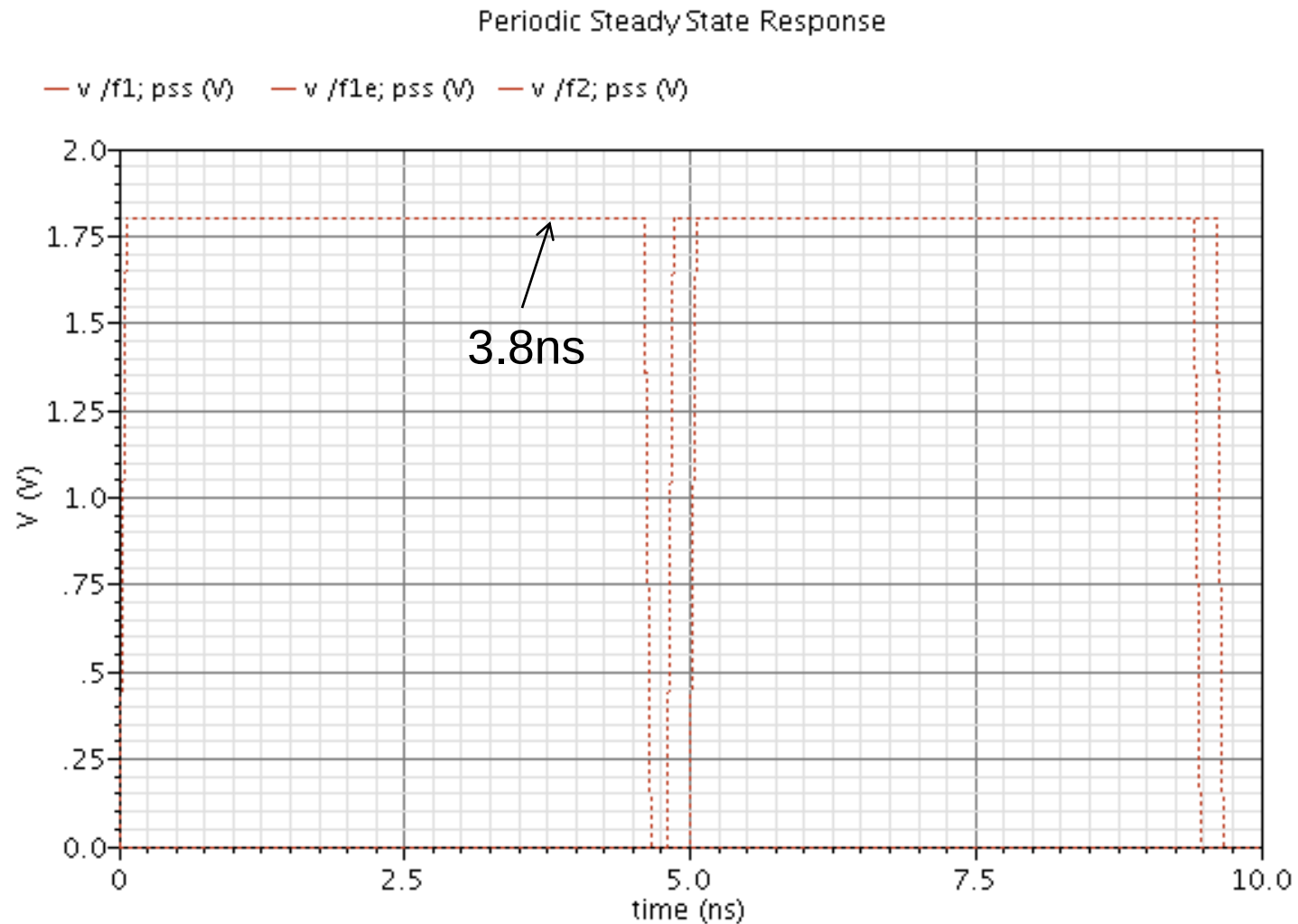
Enabled ☒ Options...

OK Cancel Defaults Apply Help

Use “tstab” if the circuit needs time to get into steady state (e.g. clock bootstrap circuits and oscillators)

Set “maxacfreq” to the highest frequency at which you expect noise to be significant (10GHz in this example; see plot on previous slide!)

PSS Waveforms (Clocks)



PNOISE Setup

PSS Beat Frequency (Hz) 100M

Sweep type **default** Sweep is Currently Absolute

Output Frequency Sweep Range (Hz)

Start-Stop **Start** 50 **Stop** 50e6

Sweep Type

Linear ☐ Step Size 40 ☐ Number of Steps

Add Specific Points ☐

Sidebands

Maximum sideband **100**

Output

voltage Positive Output Node /vcd Select

Negative Output Node /gnd! Select

Input Source

voltage Input Voltage Source /V3 Select

Reference side-band

Enter in field **0**

Noise Type **timedomain**

timedomain: strobed noise analysis

☐ Noise Skip Count

☒ Number of Points 0

Add Specific Points ☒ 3.8n

Enabled ☒ Options...

OK Cancel Defaults Apply Help

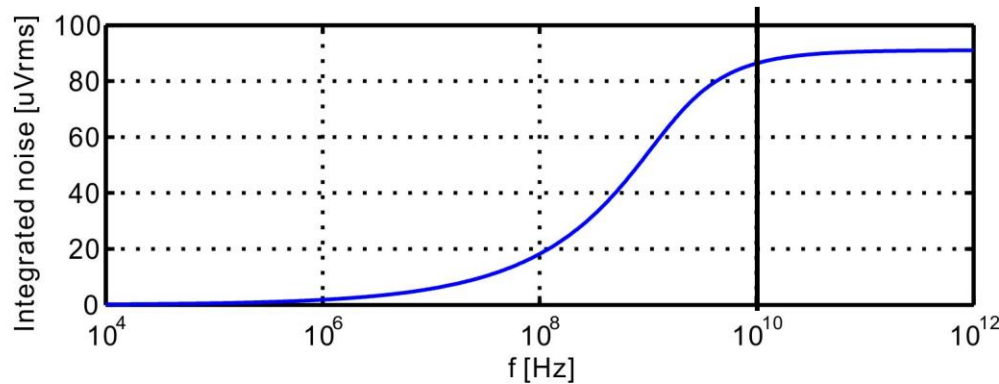
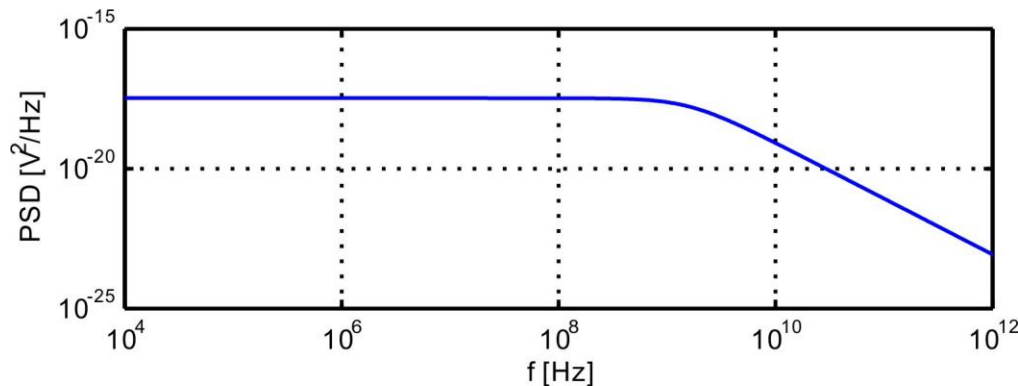
“Number of sidebands”: typically 20~200 to handle noise folding properly. Fast switches → more sidebands needed.

“timedomain” means simulator computes spectrum of discrete time noise samples

Sampling instant (3.8ns in this example)

How Many Sidebands are Needed? (1)

Sampling phase noise (.noise, $R_{on} = 100\Omega$)



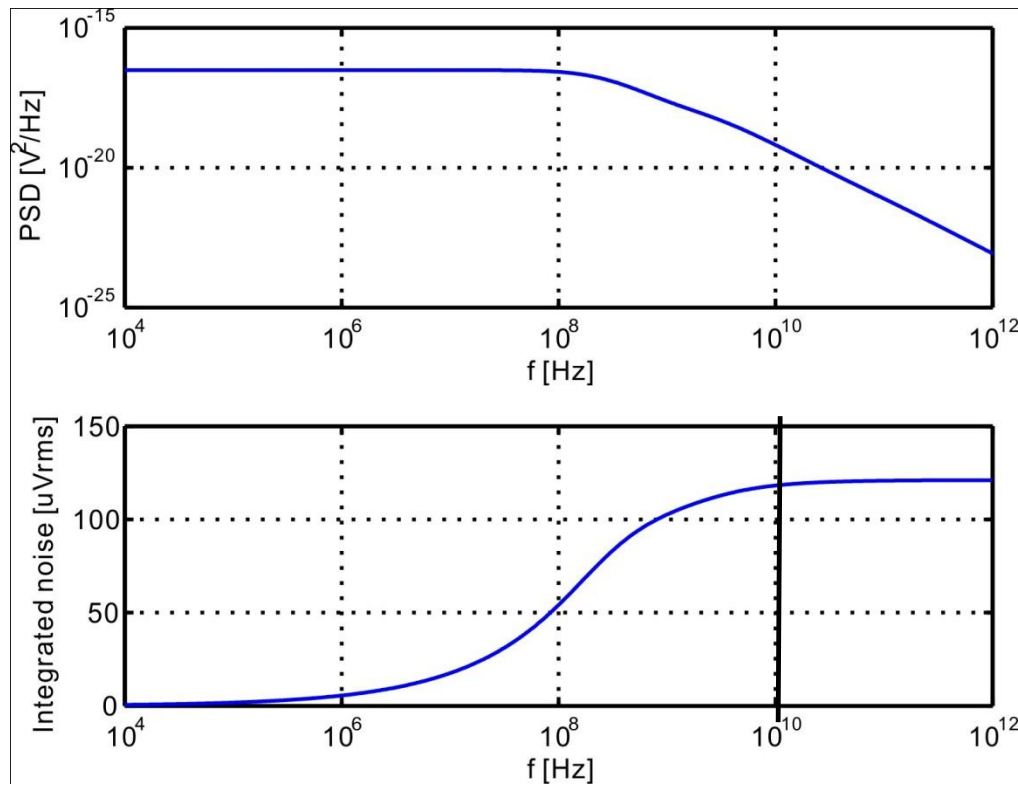
Noise up to 10GHz must be considered

→ numsidebands = 100

→ macacfreq = 10GHz

How Many Sidebands are Needed? (2)

Charge transfer noise (.noise)



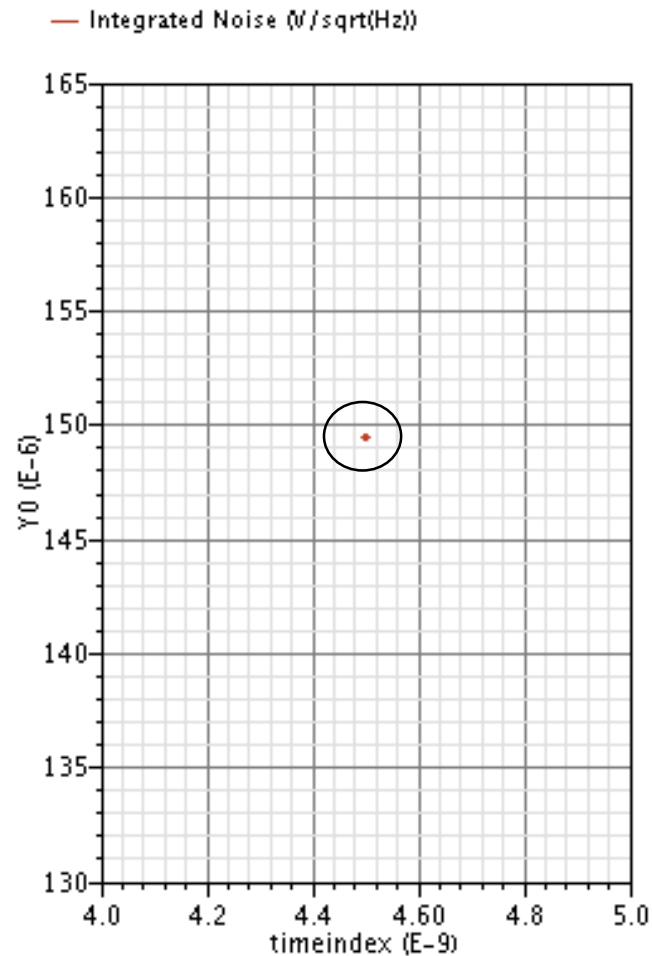
Noise up to 10GHz must be considered

→ numsidebands = 100

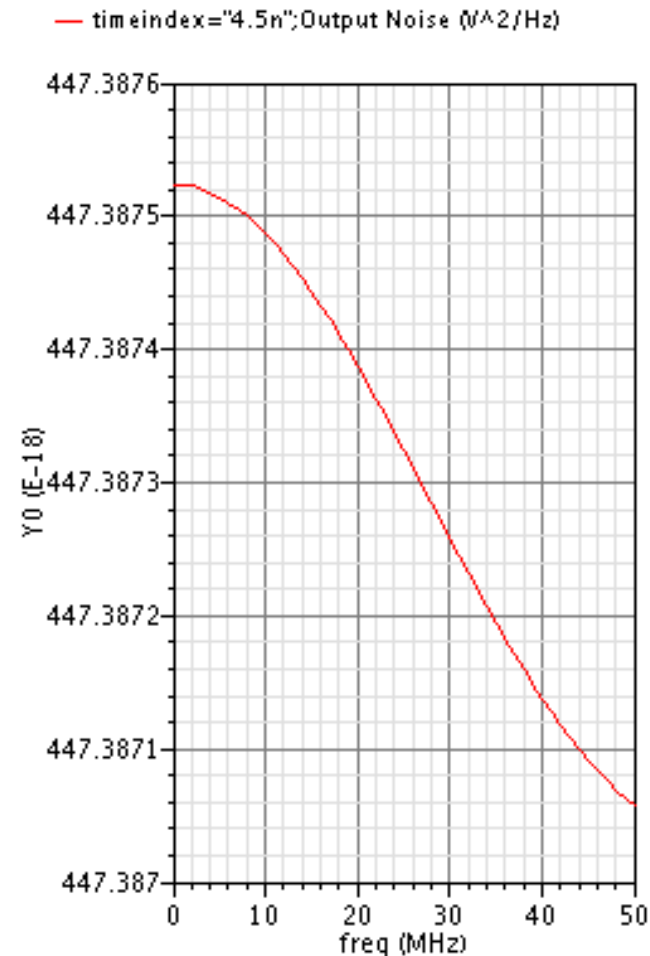
→ macacfreq = 10GHz

PNOISE Result

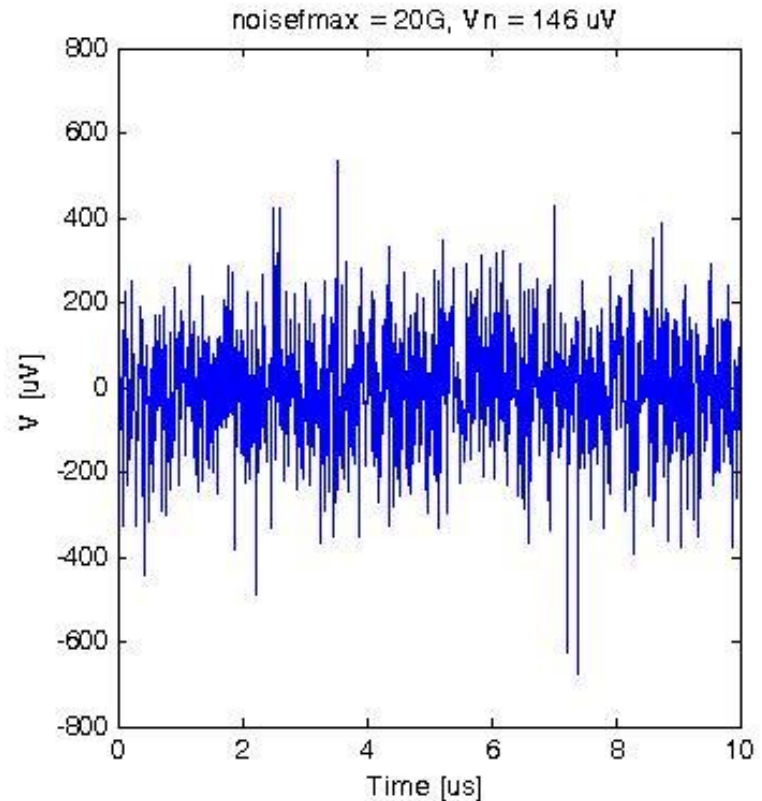
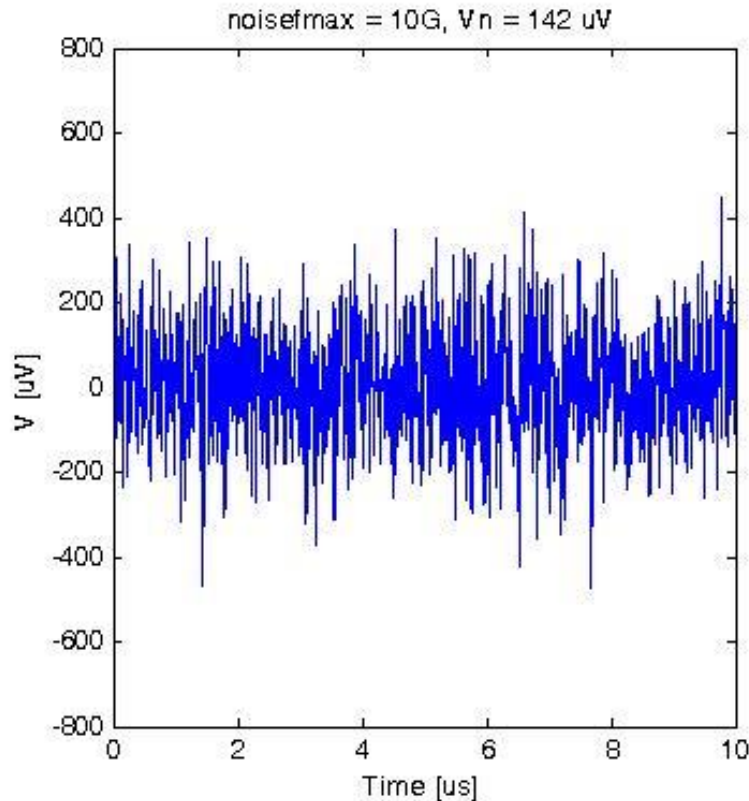
Sampled Time-Domain Noise



Noise PSD



Transient Noise



- Simulated 1000 samples
- Easy setup: only one critical parameter: noisefmax
- Larger noisefmax → more accurate result, but longer simulation time

Comparison

Method	$\phi 1$ noise [uV rms]	$\phi 2$ noise [uV rms]	Total [uV rms]	Comments
Hand calculation	91	104	138	Smaller because we have ignored the switch noise in $\phi 2$
.noise	91	121	151	
PSS/PNOISE			149	Smaller due to finite maxacfreq and numsidebands
Transient noise			146	Smaller due to max noise frequency